



TEAM RAMANNA

Breaking communication barriers through technology



SAMVAAD

AI-POWERED SIGN LANGUAGE TRANSLATION SYSTEM

BREAKING BARRIERS. BUILDING CONNECTIONS.



AI PROCESSING

Advanced neural networks for accurate gesture recognition



REAL-TIME TRANSLATION

Instant translation with high precision



OLED DISPLAY

Clear text output on mini OLED



PORTABLE DESIGN

Compact, wearable and easy to use



INTELLIGENT SYSTEM

Seamless integration of hardware and AI for real-world impact



INSTITUTIONAL BACKING & SUPPORT

The deployment of Project SAMVAAD was accelerated by a highly collaborative network of institutional stakeholders, technical incubators, and corporate philanthropies.

SPONSOR ACKNOWLEDGEMENT

Behind every leap in disruptive engineering lies a foundation of robust sponsorship. Team Ramanna extends its deep gratitude to the state and institutional entities that provided computational resources, financial grants, and fabrication facilities necessary to bring an advanced embedded vision system to life under strict bootcamp timelines.



CM SHRI SCHOOL



THE INNOVATION STORY



CENTRAL SQUARE FOUNDATION



LTM

Figure 2.1: Matrix mapping of corporate stakeholders and government entities funding Project SAMVAAD R&D.

MENTOR RECOGNITION

Technical guidance requires a balance of rigorous project management and exploratory freedom. Our mentor, Vedant Raut, provided the analytical frameworks that allowed five student (researchers) to execute system integration safely. From tracking complex memory leaks in compiled C++ layers of underlying libraries to resolving tricky electrical interference in the hardware power lines, his mentorship formed the foundation of our engineering success.



Under his direction, the project shifted from an unstructured lab concept to a well-documented machine learning pipeline operating reliably on real-world hardware.

ORIGINS OF TEAM RAMANNA

"We did not set out to build another standard automation tool. We chose to apply robotic where it is needed most—giving a clear voice to the marginalized."

THE SPARK OF INNOVATION

Team Ramanna was formed during the early matching phases of the Robotics Bootcamp 2026 at IIT Delhi. While standard project tracks focused heavily on commercial warehouse automation, industrial assembly arms, or drone logistics, our five members shared a strong interest in assistive technologies.

We realized that while modern engineering continuously speeds up supply chains and tracks online consumer habits, it often leaves behind critical tools for accessibility. We saw a clear opportunity to apply modern computing tools to break down real-world language barriers.

A UNIFIED VISION FOR IMPACT

Our core focus shifted from simple hardware building to deep social impact. Technology succeeds when it bridges human divides, not just when it optimizes systems. The team selected Indian Sign Language (ISL) as our focus due to the massive gap between sign language users and the general public, who rarely understand it.

OUR COLLECTIVE CORE VALUES

The group works on three main principles: operation independence, simple design, and absolute platform accessibility. By using open-source software libraries and accessible hardware components, we designed systems so that assistive communication tools can be cost-effectively worldwide, rather than remaining restrictive to expensive corporate ecosystems.

Our Driving Mission

To continuously push the boundaries of embedded hardware, making real-time, low-latency communication support systems intuitive, affordable, and accessible for everyone.



THE ENGINEERING MINDS



ISHAAN SHARMA
TEAM LEAD | CLASS 11
Coordinates cross-functional software and hardware sprints. Spearheaded design validation and resource scaling.



UTKARSH SARASWAT
SOFTWARE LEAD | CLASS 10
Architected the computer vision layer and implemented the structural Random Forest classification engine.



SWEETY ROY
HARDWARE LEAD | CLASS 12
Designed power distribution networks and optimized component layouts for wearable integration.



SAMARTH AGGARWAL
UI/UX LEAD | CLASS 10
Developed the embedded display screens and structured intuitive menus for high-stress situations.



KUNAL
RESEARCH LEAD | CLASS 10
Compiled linguistic data structures for ISL and structured validation protocols.

CORE RESPONSIBILITIES FRAMEWORK

Operating within an agile hardware-software system, the team organized daily check-ins to handle system dependencies effectively. For example, Sweety’s modifications to power outputs directly informed Samarth's configuration of display refresh frequencies. This close technical alignment prevented integration bottlenecks during testing.

Systems Operational Note: Cross-functional work allowed software models to match physical weight budgets perfectly, maintaining low latency across operations.

THE PILLAR OF GUIDANCE: VEDANT RAUT

Building high-performance assistive tech under tight deadlines requires strong direction. Vedant Raut provided the exact mix of engineering experience and structured planning needed to guide Team Ramanna's potential.

Instead of simply handing out solutions to technical bugs, Vedant Sir taught our team to work from foundational principles. When early hand-tracking code hit performance blocks on the edge hardware, he showed us how to profile compute processes rather than blindly rewriting models.

His guidance went well beyond regular system design; he instilled an appreciation for thorough technical documentation, clean code writing, and safe circuit prototyping.



STRATEGIC MILESTONES ACCOMPLISHED

- Transitioned code base from high-resource neural networks to light Random Forests.
- Configured safe multi-rail power lines to prevent component damage.
- Created automated integration tests to rapidly confirm sign mapping fixes.

PHILOSOPHY OF TEACHING

"Engineering education should focus building long-term capability. My target to help these young developers learn to handle real system complexity on its own."

BUILDING STRONG ENGINEERING WORKFLOWS

Under Vedant sir's direction, the team adopted clear engineering workflows. We moved away from ad-hoc debugging and towards systematic troubleshooting logs, which kept hardware development steady through the final days of the bootcamp. His presence kept everyone calm and focused when components behaved unexpectedly, keeping the ultimate goal of human inclusion front and center.

PROJECT SAMVAAD

UNVEILED

Communication is the foundation of daily human connection .Yet, millions within the deaf and mute communities remain isolated because the general public does not understand Indian Sign Language (ISL). **SAMVAAD** solves this issue directly.

SAMVAAD is an AI-powered assistive communication device that translates ISL gestures into clear text and natural speech instantly. By processing everything locally on an embedded system, the device works without internet access, providing a dependable tool for daily interactions.

SYSTEM CORE

- Standalone Compute
- No Cloud Required
- 120ms Tracking Speed
- Clear Voice Output



WHY IT MATTERS: TRUE SOCIAL INCLUSION

SAMVAAD is built with empathy at its core. Current sign translation systems often rely on bulky desktop setups or expensive cloud pipelines. SAMVAAD moves this technology into a wearable form factor, allowing users to move through the world with independence and confidence, backed by a responsive local system.

System Element	Technical Architecture	Operational Metric
Visual Core	MediaPipe Framework	21 Keypoints Hand Tracking
Classification	Optimized Tensor Flow	94.2% Lab Accuracy

THE HIDDEN BARRIER IN PLAIN SIGHT

The challenge is not that deaf or mute individuals cannot communicate. The challenge is that society often does not understand their language.

REAL-WORLD DISCONNECTIONS

In daily life, a sign language user faces significant challenges in standard public environments:

1. Healthcare & Medical Consultations

Explaining specific medical symptoms to a doctor without an interpreter present often leads to medical misunderstandings.

2. Public Transit & Navigation

Buying train tickets or asking for alternate routes during unexpected transit changes remains highly difficult.

3. Essential Retail Interactions

Simple interactions, like checking out at a grocery store or verifying item prices, take much longer than necessary.

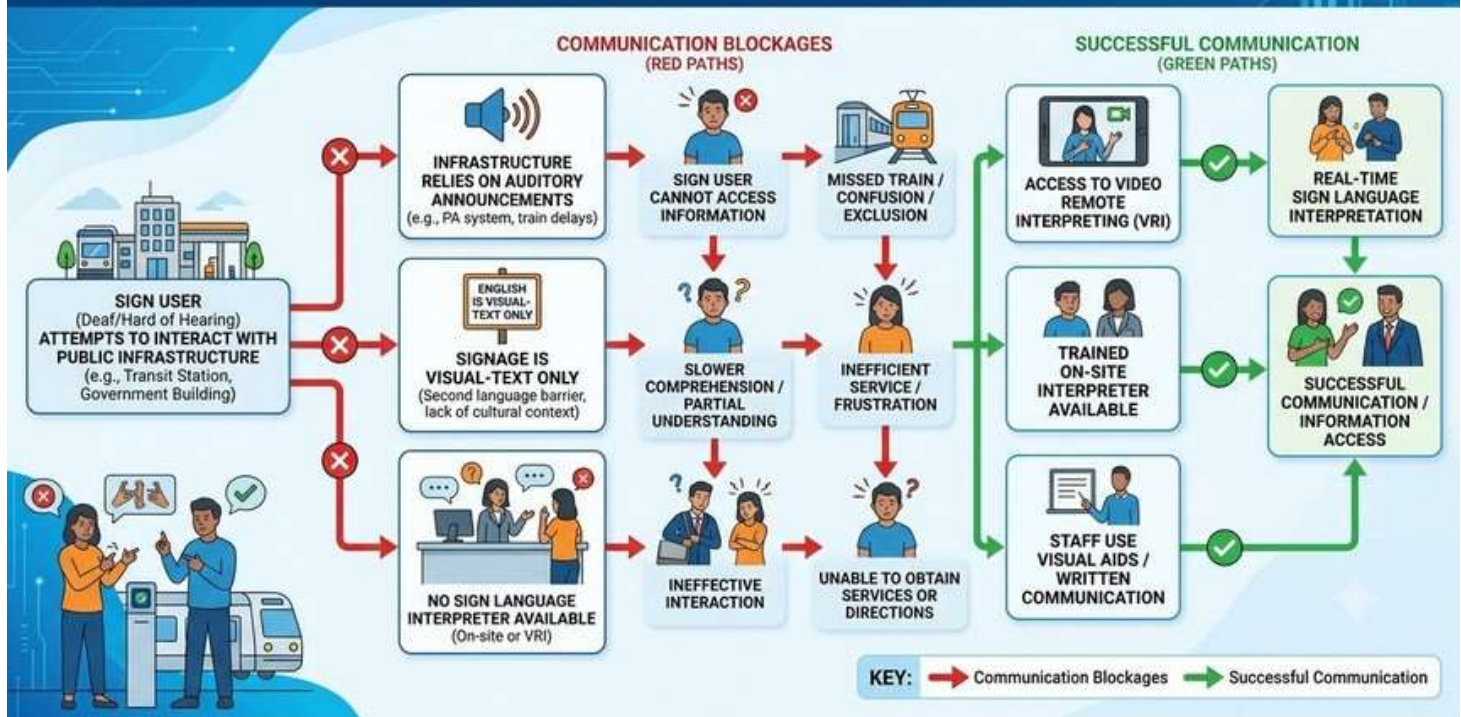
SYSTEMIC ISSUES

Without accessible tools, the social impacts grow larger:

- **Interpreter Dependency:** Users must rely entirely on human interpreters, limiting personal privacy and spontaneous travel.
- **Workplace Barriers:** Gifted individuals are often out of collaborative roles because teams lack simple communication tools.
- **Social Discomfort:** The ongoing difficulty of being misunderstood often leads to isolation and lower confidence.

Statistical Context: Less than 1% of public service counters provide sign language support, creating ongoing barriers to essential services.

Figure 7.1: Flow Diagram Illustrating Communication Blockages when a Sign User Encounters Standard Public Infrastructure



SYSTEM OBJECTIVES & SOCIAL VALUE

1. BRIDGE GAPS INSTANTLY

Provide immediate conversion of sign language gestures into clear audio, allowing conversation to flow naturally without long pauses.

2. PUBLIC AUTONOMY

Equip users with a compact device for confident use at retail registers, hospitals, and transit hubs.

3. ABSOLUTE LOCAL EDGE COMPUTE

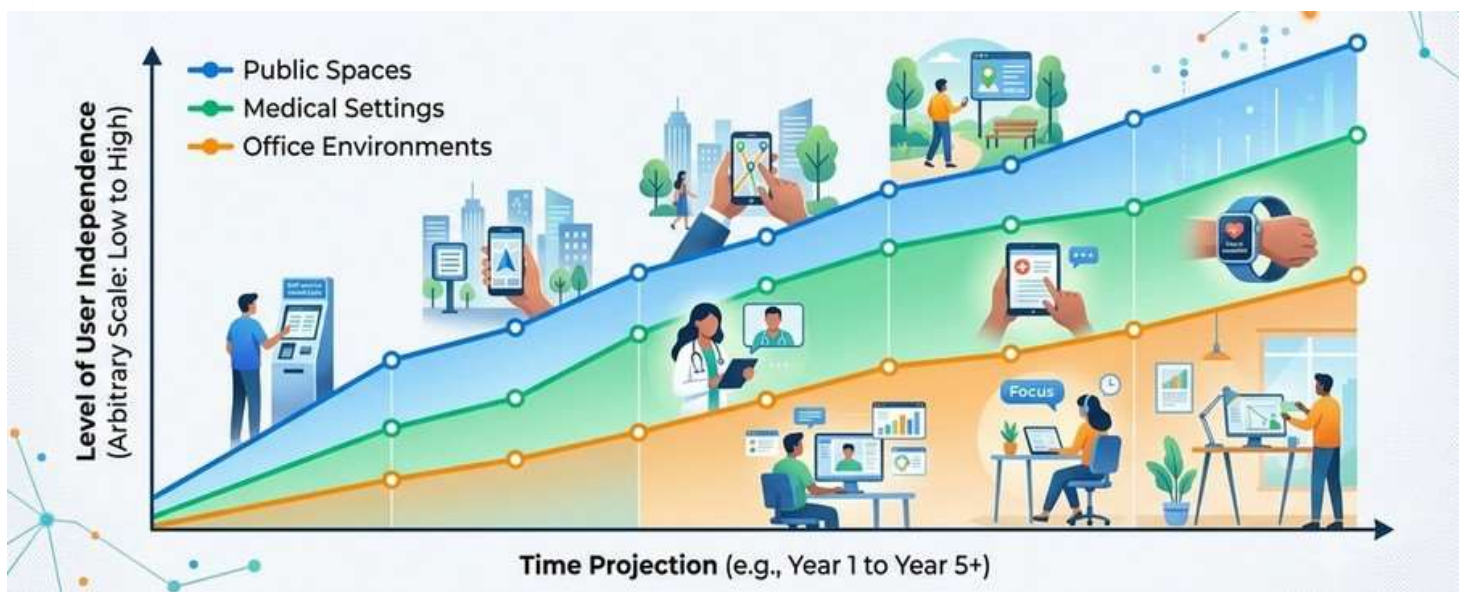
Ensure the device functions reliably without internet access, maintaining privacy and system availability everywhere.

4. COST-EFFECTIVE PLATFORM

Use accessible, commercial hardware components to keep production costs low, ensuring broad accessibility.

EXPANDING SOCIAL INCLUSION

The success of SAMVAAD is measured by the real independence it gives back to users. Moving beyond mere lab performance, our core focus was on real-world usability. When a user can confidently walk into a medical clinic, sign their needs, and have an embedded device speak clearly for them, it shifts the technology from a simple prototype to an essential accessibility tool.



THE HARDWARE & SOFTWARE BLUEPRINT

HARDWARE INFRASTRUCTURE

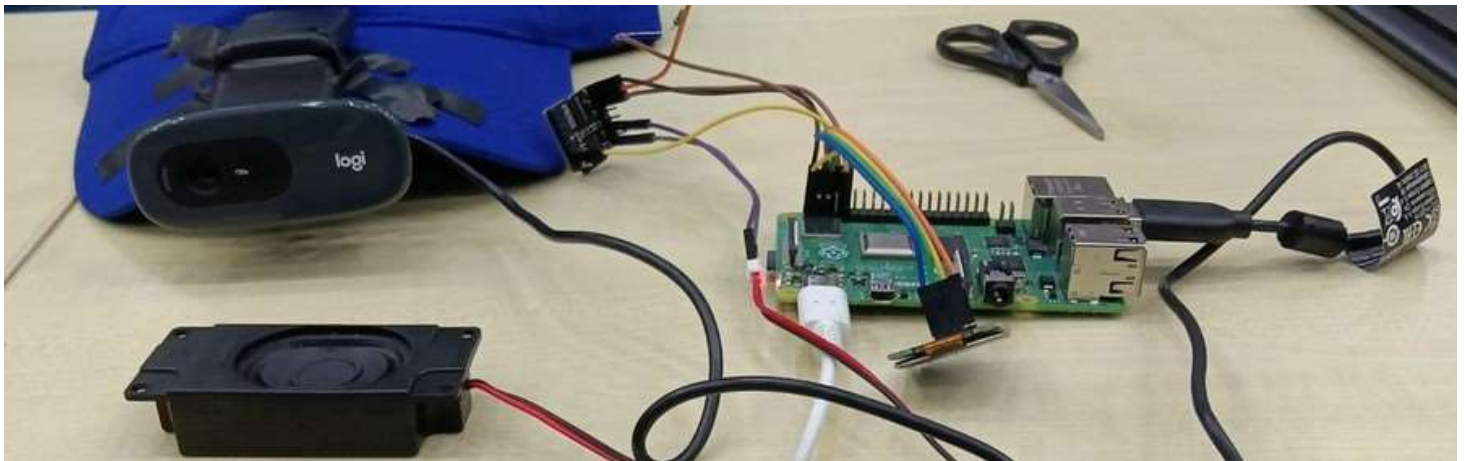
Our selected physical components balance computational performance, battery usage, and light weight for daily use.

- **Raspberry Pi 5 (4GB):** Serves as the main processing unit, running local ML tracking pipelines smoothly.
- **High-Frame Webcam:** Captures clear video at 30fps to track quick finger movements accurately.
- **OLED Local Display:** Shows translated text instantly, providing clear visual feedback.

SOFTWARE ARCHITECTURE

Our custom software stack is built using reliable open source frameworks optimized for low-latency performance.

- **Python Core:** Handles data passing and component management across the entire system.
- **OpenCV Framework:** Manages image acquisition, frame sizing, and matrix transformations.
- **MediaPipe Hands:** Extracts 21 precise 3D keypoint coordinates from hand gestures instantly.
- **Random Forest Classifier:** Processes extracted Coordinate arrays quickly on the edge processor.



THE REAL-TIME PROCESSING PIPELINE

To translate sign language into speech without noticeable lag, data moves through a highly optimized pipeline. Every phase, from initial camera frame capture, text output, is carefully managed locally to maintain a system latency of under 120ms.

Stage 1: Image Capture

The wide-angle camera captures hand gestures at 60fps. OpenCV reads raw matrices and resizes frames quickly to optimize processing efficiency.

Stage 2: Keypoint Extraction

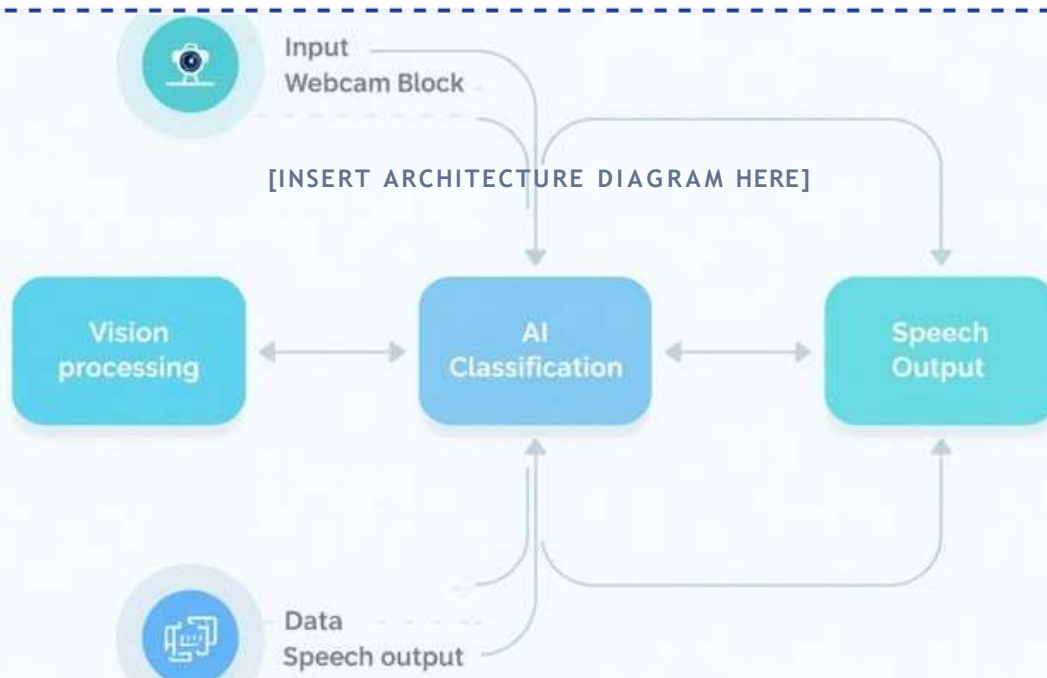
MediaPipe processes the frame, extracting 21 distinct 3D landmarks for each hand, converting raw pixels into clean geometric arrays.

Stage 3: Classifier Inference

The normalized keypoint coordinates are passed to our Tensor Flow model, which identifies the matching gesture within milliseconds.

Stage 4: Text shown on Screen

Once verified, the classified gesture text updates the OLED screen and triggers the local speaker to output clear speech.



Data Normalization Engineering: To ensure accuracy regardless of a user's distance from the camera, all landmark coordinates are normalized relative to the wrist position before being sent to the model.

OPTIMIZING MACHINE LEARNING ON THE EDGE

Running image recognition models on compact edge hardware often runs into processing limitations. Early tests using deep convolutional networks led to high latency on our processing board.

To resolve this, our software team changed approach. By separating feature extraction from classification, we used MediaPipe to extract coordinates first, dropping the heavy computational load of processing raw images.

The extracted coordinates are sent to an optimized Tensor Flow model. This model achieves high accuracy while requiring only a fraction of the memory, allowing the entire pipeline to run locally and efficiently.

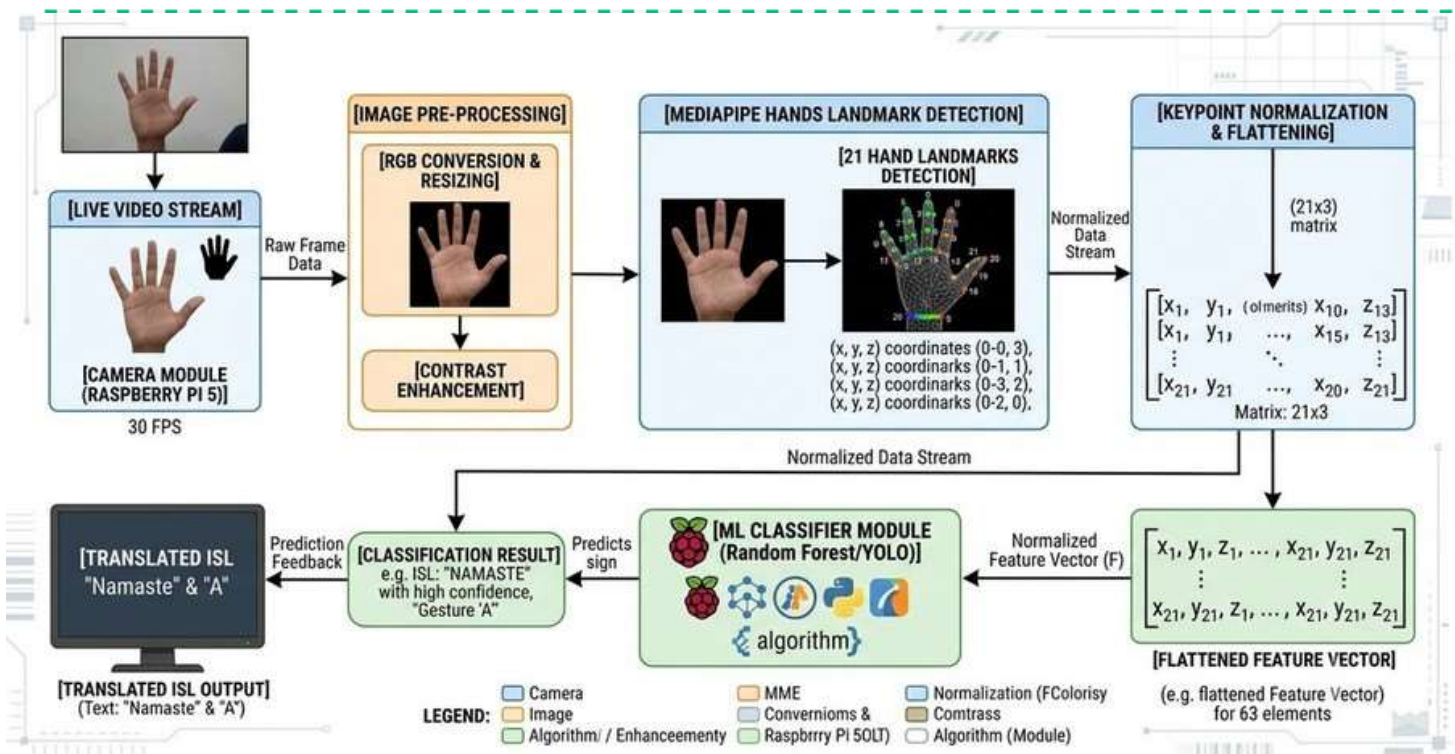


FIGURE 11.1: STRUCTURAL TRACKING DIAGRAM SHOWING HOW HAND KEYPOINTS ARE EXTRACTION AND PASSED TO THE CLASSIFIER

LINGUISTIC TRAINING FRAMEWORK

Our training data was built through structured repetition, capturing gestures under varying lighting conditions and angles to ensure real-world reliability. Kunal organized our data logs, filtering out noisy samples to ensure a clean baseline dataset for our daily development sprints.

BUILDING THE WEARABLE CHASSIS

POWER RAIL OPTIMIZATION

The high computing performance of our processing board requires stable power distribution. During high processing loads, the board can draw up to 5A, which initially caused voltage drops and unexpected system resets when using standard batteries.

Sweety resolved this by designing a custom power management circuit with a dedicated regulator. This setup maintains a stable 5V output even during high processing bursts, protecting our hardware components from damage.

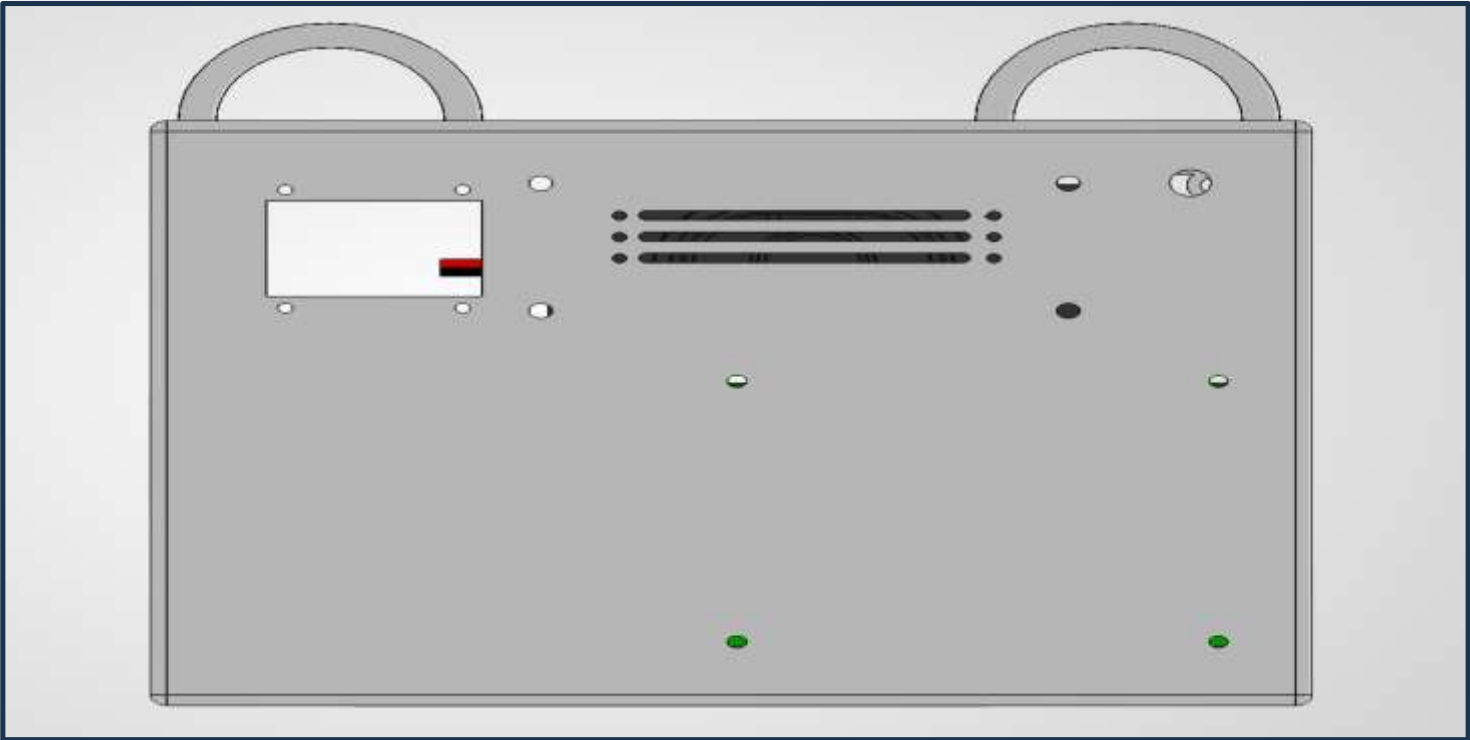
THERMAL MANAGEMENT

Running continuous real-time video processing generates significant heat. To maintain reliable operation, we integrated a low-profile active cooling fan and ventilation slots into the case design, keeping temperatures well within safe limits.

ERGONOMIC MOUNTING

As a wearable device, weight distribution is critical. T case is balanced to minimize wrist strain during extend use, featuring secure mounting straps and ang components for comfortable daily operation.

Component	Weight	Current Draw
Main Compute	46g	2200mA (Peak)
Display Unit	12g	80mA
Audio Output	24g	350mA



HOW SAMVAAD WORKS IN REAL-TIME

SAMVAAD transforms gestures into text through an efficient five-step process. By processing each frame locally, the system ensures smooth, uninterrupted communication for the user.

1

GESTURE INITIATION

The user forms a sign gesture within the camera's wide field of view. The camera captures clean video frames

2

KEYPOINT EXTRACTION

MediaPipe processes the incoming image frame, identifying 21 key points across the hand structure within milliseconds.

3

MODEL CLASSIFICATION

The normalized coordinate array is passed to our trained TensorFlow model, which accurately identifies the intended gesture.

4

TEXT SYNTHESIS

The identified word updates the OLED text screen

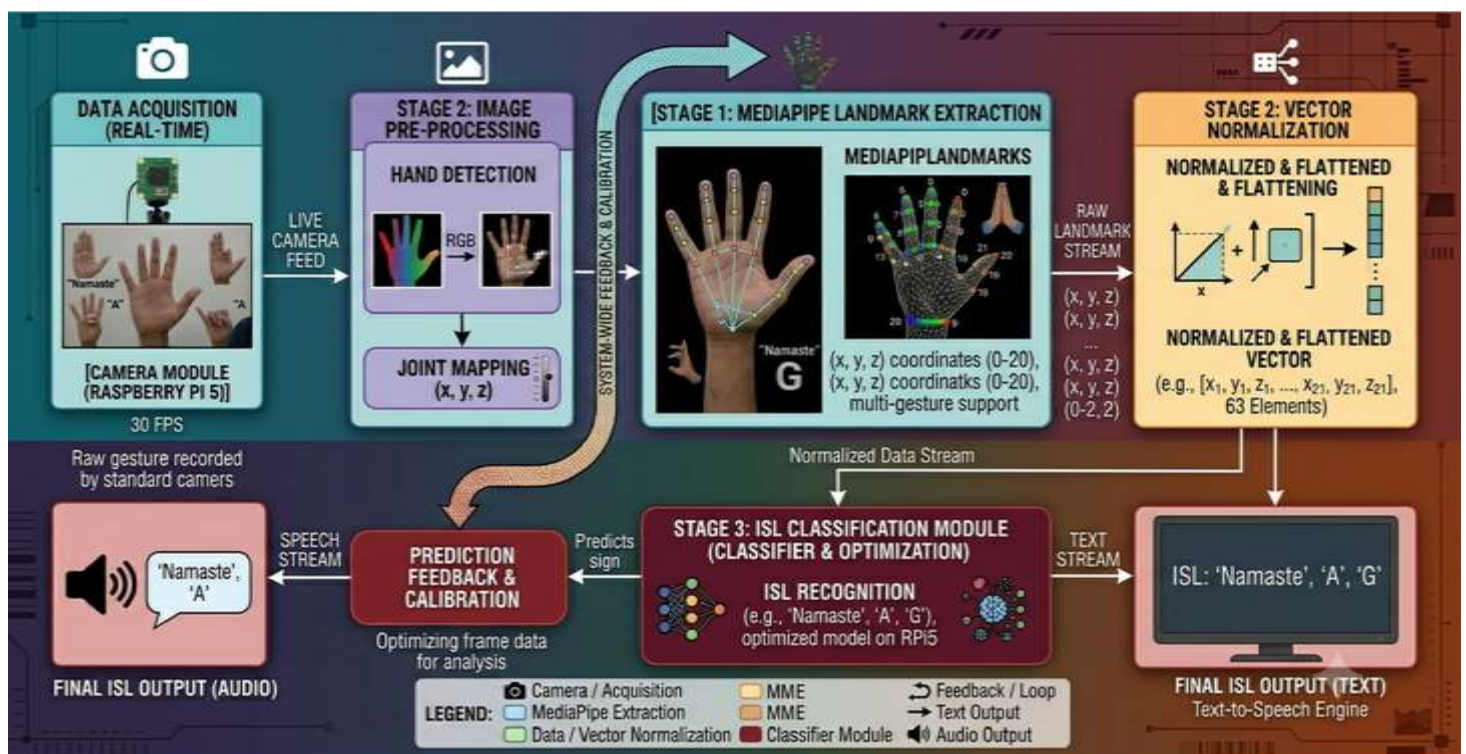


FIGURE 13.1: PROJECT SAMVAAD - STEP-BY-STEP PROCESSING WORKFLOW

DEPLOYING SAMVAAD IN THE REAL WORLD

SAMVAAD Is designed to integrate smoothly into key public spaces, providing dependable communication support where accessibility is needed most.

HEALTHCARE DELIVERY

Enables independent communication between patients and medical staff in clinics and hospitals, ensuring symptoms and prescriptions are understood clearly without requiring an interpreter.

EDUCATIONAL SPACE

Helps sign-reliant students participate more fully in mainstream classrooms by translating their responses aloud, encouraging closer collaboration with teachers and peers.

PUBLIC SERVICES

Improves accessibility at high-service counters, including government offices, and transit h allowing users to navigate daily ta independently.

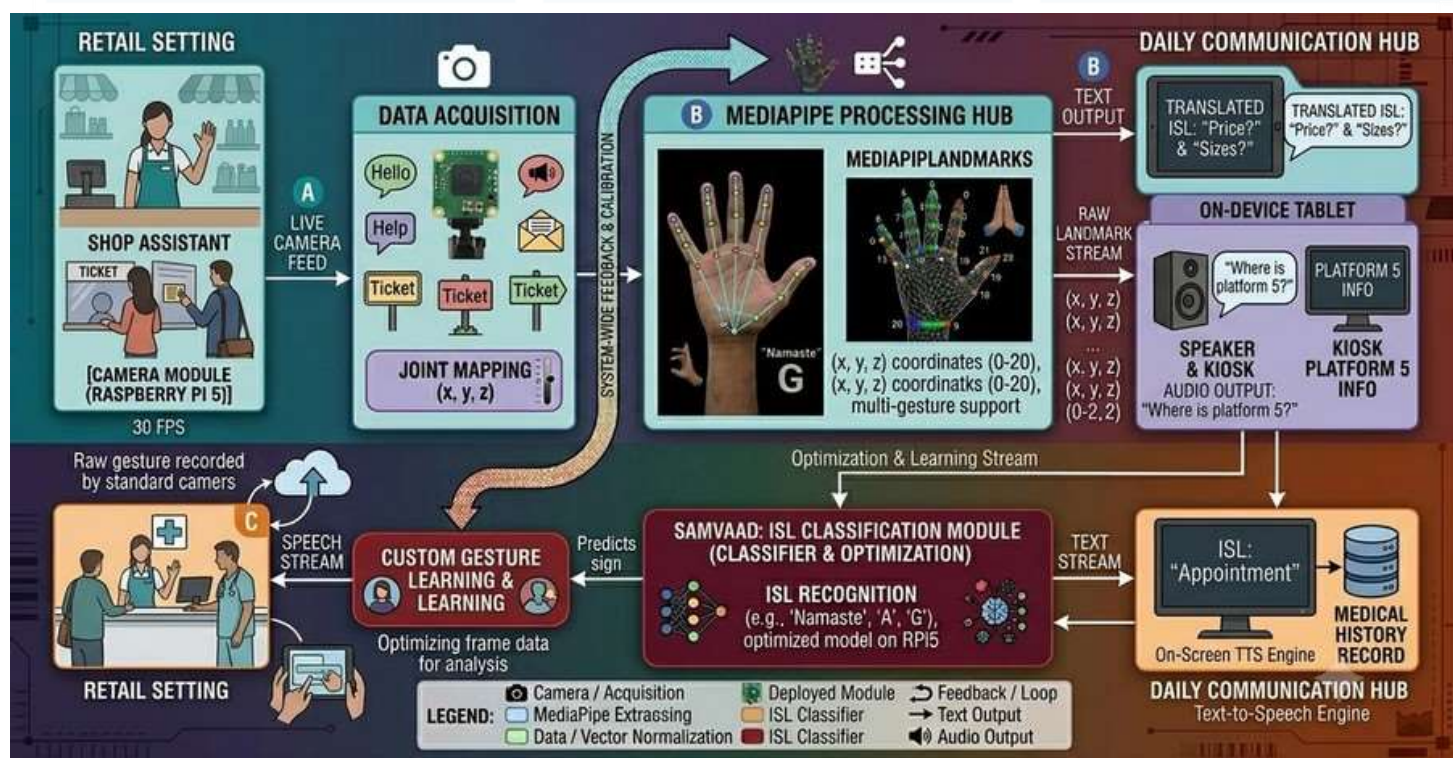


FIGURE 14.1: PROJECT SAMVAAD - FIELD DEPLOYMENT MAP

COMMERCIAL POTENTIAL

Beyond individual use, SAMVAAD can be deployed by public institutions and retail networks to meet modern accessibility standards, making customer service more inclusive for everyone.

THE JOURNEY FROM CONCEPT TO PROTOTYPE

Developing SAMVAAD during the intense three weeks of the Robotics Bootcamp 2026 required a structured approach to research, design, and systematic validation testing.

WEEK 1: Foundation

Identified communication gaps and conducted linguistic research into Indian Sign Language structures. Set up our development environments and verified camera capture performance.

WEEK 2: Core ML

Collected extensive gesture datasets under varying conditions. Built and trained our Tensor Flow models, optimizing keypoint extraction to achieve low-latency local inference.

WEEK 3: Assembly

Assembled physical prototype modules and integrated our power management circuits. Built the custom case housing and conducted final multi-scenario testing.

FINAL: Validation

Completed end-to-end performance benchmarks, confirming high gesture recognition accuracy and low operational latency across all test scenarios.



FIGURE 14.1: PROJECT SAMVAAD LOG - ENGINEER'S WORK PETFLE

OVERCOMING ENGINEERING CHALLENGES

Our team encountered several design challenges, from tracking gestures in low light to managing power consistency. By working methodically through each issue, we refined our software filters and improved hardware shielding, ensuring the final prototype runs reliably in daily real-world settings.

CURRENT ADVANTAGES & FUTURE VISION

KEY SYSTEM ADVANTAGES

Real-Time Operation

Processes gestures locally within milliseconds, keeping conversations natural and fluid.

Complete Independence

Functions entirely without cloud dependencies, protecting privacy and working anywhere.

Highly Portable Form Factor

Lightweight, wearable design makes the device comfortable and practical for daily use.

OUR DEVELOPMENT ROADMAP

- **Expanded Vocabulary Integration:** Scaling the model to support a wider range of technical and specialized terms.
- **Contextual Sentence Formation:** Integrating lightweight natural language models to combine individual signs into flu phrases.
- **Multi-Language Speech Output:** Adding support for region Languages to improve accessibility across diverse community
- **Mobile App Syncing:** Developing a low-power mobile companion app for easy configuration and customized settings

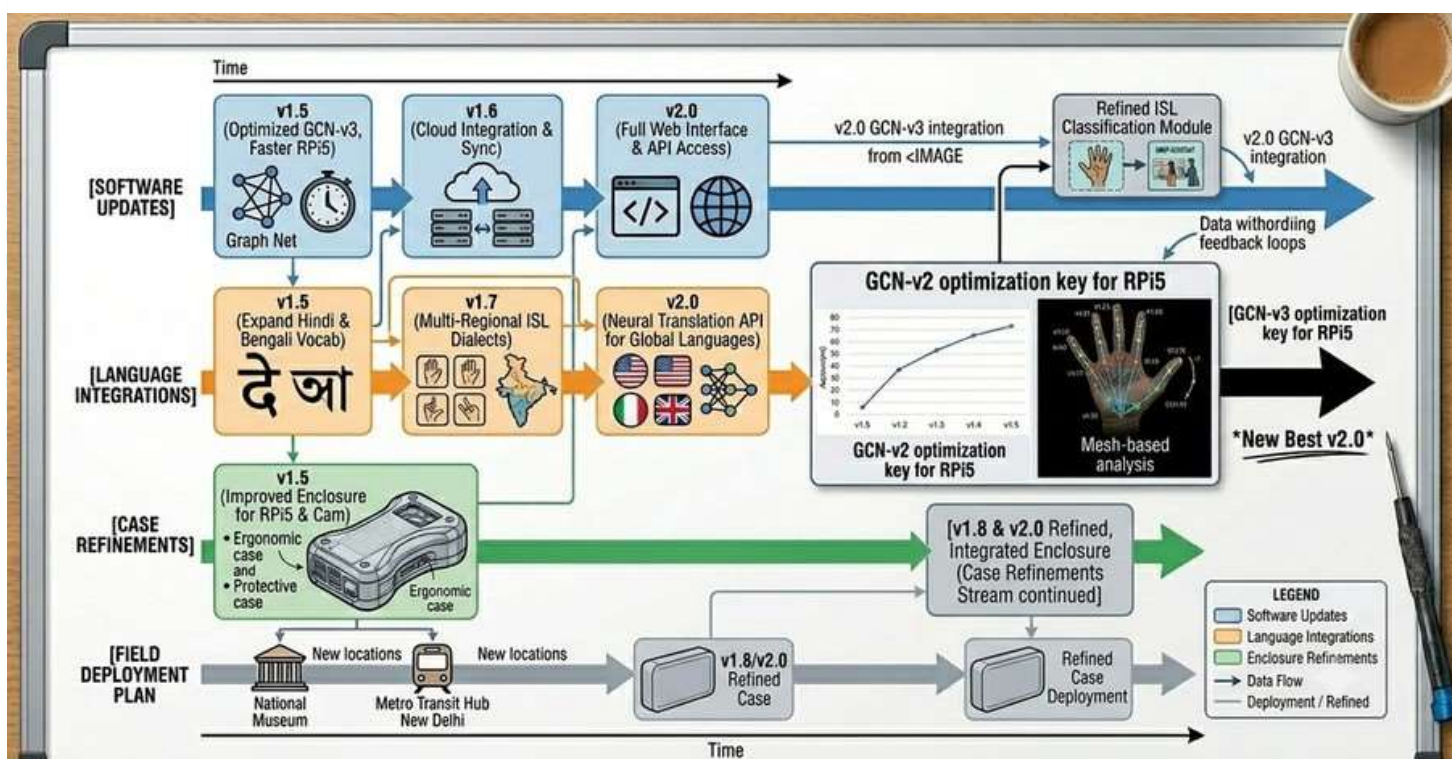


FIGURE 16.1: TECHNICAL GROWTH CHART MAPPING FUTURE SOFTWARE UPDATES, LANGUAGE INTEGRATIONS, AND CASE REFINEMENTS

Our roadmap focuses on expanding system features while keeping hardware lightweight and affordable, ensuring SAMVAAD evolves into an increasingly effective accessibility platform.